# ADARSH KUMAR SINGH

**Full-Stack Software Engineer** | React · Node.js · Python/FastAPI · AWS · AI-Assisted Development

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## PROFESSIONAL SUMMARY

Full-Stack Software Engineer (B.Tech CSE, 2027) who ships end-to-end web applications: React/JavaScript frontends, Node.js and FastAPI REST APIs, WebSocket real-time services, SQL/NoSQL databases, and AWS. Experienced in authentication (OAuth 2.0, RBAC), security hardening, WCAG 2.2 accessibility, and automated testing (276 tests).

## EDUCATION

**Vellore Institute of Technology** 2023 – 2027

*B.Tech, Computer Science and Engineering – CGPA: 8.28/10* *Bhopal, Madhya Pradesh*

- **Relevant Coursework:** Data Structures & Algorithms, OOP, Operating Systems, DBMS, Computer Networks, Cloud Computing

## EXPERIENCE

**Software Engineering Intern** May 2026 – Jul 2026

*MPOnline Limited – TCS & Govt. of Madhya Pradesh Joint Venture* *Bhopal, Madhya Pradesh*

- Built **Lexicon Library**, a full-stack library management app with passwordless email two-factor auth, book/borrow CRUD, search, bulk actions, CSV export, a Chart.js analytics dashboard, and an auto-recalculated overdue-fines engine.
- Architected a Cloudflare Worker proxy that keeps the Brevo email API key server-side and validates requests via Origin/header checks, preventing client-side secret exposure.

## PROJECTS

**VoxBuddy** – Real-Time Voice AI App | *Python, FastAPI, WebSockets, AWS, Redis, Gemini* Sep 2026

- Built a real-time FastAPI/WebSocket pipeline streaming 16 kHz audio through speech recognition, Gemini translation, and Amazon Polly TTS with async processing and automatic reconnection; deployed on AWS Elastic Beanstalk with DynamoDB, Redis-backed sessions, OTP/Google sign-in, and an installable PWA (Capacitor Android/iOS). Designed provider-independent architecture (Python Protocols, config-driven factories) that switched translation from Bedrock to Gemini by configuration alone.
- Fixed a cross-user authorization flaw and a transaction-commit race condition found in live testing; wrote 276 Pytest tests (AWS mocked with moto) that caught a Polly voice-ID bug. Used Claude and ChatGPT for scaffolding and debugging, verifying all output through tests.

**SenseRelay** (Live Demo) – Biosignal Monitoring | *Python, FastAPI, SQLite, React, TypeScript* 2026

- Built a synthetic biosignal-monitoring system that validates TCP newline-delimited JSON with CRC-32 checks and durably records samples, data-loss events, and upload state in SQLite.
- Implemented idempotent authenticated ingestion, retryable sync after outages, and a React dashboard for sessions, results, missing data, and service status; 31 Python tests pass.

**CampusVibe** (Live Demo) – Event Management Platform | *JavaScript, Node.js, Express.js, SQLite, JWT, Google OAuth 2.0, WCAG 2.2* Jul 2026

- Built a full-stack event platform (12 pages, REST APIs) with RBAC, QR e-tickets, discount codes, attendance scanning, payment verification, analytics, and a 5-minute seat hold that releases unpaid capacity.
- Ran a security audit that fixed IDOR/BOLA access-control gaps (including Critical/High findings); resolved 19 WCAG 2.2 issues; hardened 20+ async workflows with timeouts, retries, and stale-while-revalidate caching; supports 32 languages (i18n).

**CallBeacon** – Missed-Call Auto-Reply App | *Java, Android SDK, SQLite* Nov 2025

- Identified a common user problem – missed calls going unanswered with no follow-up – and designed configurable, scheduled SMS auto-replies so users control when and how they follow up.
- Added delivery tracking to validate that replies reach contacts (98% delivered across 50 test messages) and built a companion website for distribution.
- Tracked website analytics as an early reach signal: 71 visits since Jul 8 from 6+ countries (India 31, US 6, France 5, Netherlands 5), 80% on Android (matching the Android-first product), and 18 APK-download events.

## TECHNICAL SKILLS

**Languages:** JavaScript, TypeScript, Python, Java, C++, SQL, HTML5, CSS3
**Frontend:** React, Responsive Design, WCAG 2.2 Accessibility, i18n, Chart.js
**Backend & APIs:** Node.js, Express.js, FastAPI, REST APIs, WebSockets, Asynchronous Processing, Prisma
**Databases:** PostgreSQL, SQLite, DynamoDB (NoSQL), Redis
**Cloud & DevOps:** AWS (Elastic Beanstalk, DynamoDB, Polly, S3, CloudWatch), Azure, Cloudflare Workers, Git, CI Basics
**Security & Testing:** OAuth 2.0, JWT, RBAC, OTP Authentication, IDOR/BOLA Remediation, Pytest, Postman, Debugging
**AI & AI-Assisted Development:** Claude, ChatGPT, LLM API Integration (Gemini, OpenAI), Prompt Engineering, AI Agents, scikit-learn

## ACHIEVEMENTS & CERTIFICATIONS

- Global Rank 11,779 of 146,922 in TCS CodeVita Season 13 (top 8%) for algorithmic problem-solving.
- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate – 98% exam score (Dec 2025).
- Google's 5-Day AI Agents: Intensive Vibe Coding Course (2026) – capstone in AI-assisted software development.
- Software Engineering Job Simulation, Electronic Arts (Forage) – feature development, OOP, live bug fixing.